

Advanced Microeconomics

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Problem Set 3: Mathematical solutions

2. Application

1) There is an individual with utility function $U(x_1, x_2) = \sqrt{x_1} + x_2$. Wealth is given by $w > 0$ and price are $p_1, p_2 > 0$.

a) Determine the optimal consumption, the indirect utility function and the expenditure function.

Solution:

$$x_1(p_1, p_2, w) = \frac{1}{4} \left(\frac{p_2}{p_1} \right)^2$$

$$x_2(p_1, p_2, w) = \frac{w}{p_2} - \frac{p_2}{4p_1}$$

$$v(p_1, p_2, w) = \frac{w}{p_2} + \frac{p_2}{4p_1}$$

$$e(p_1, p_2, u) = p_2 u - \frac{p_2^2}{4p_1}$$

b) Assume that the prices are given by $p_1 = p_2 = 3$. The income is $w = 120$ monetary units. Calculate the demanded quantities x_1 and x_2 and the corresponding utility level.

Solution:

$$x_1 = 0,25$$

$$x_2 = 39,75$$

$$u = 40,25$$

2) The expenditure function of an individual is given by $e(p, u) = u\sqrt{p_1 p_2}$. Accordingly, there are two consumption goods whose consumption quantities are denoted by x_1 and x_2 and whose prices are denoted by p_1 and p_2 , respectively.

a) Calculate the indirect utility function.

Solution:

$$v(p_1, p_2, w) = \frac{w}{\sqrt{p_1 p_2}}$$

- b) Determine how much of x_1 and x_2 the individual consumes in the optimum as a function of prices and income.

Solution:

$$x_1(p_1, p_2, w) = \frac{w}{2p_1}$$

$$x_2(p_1, p_2, w) = \frac{w}{2p_2}$$

- c) Assume that the parameters are determined by the following values $(p_1^0, p_2^0, w^0) = (3, 3, 120)$ Determine the quantities demanded and the associated utility level.

Solution:

$$x_1 = 20$$

$$x_2 = 20$$